Sequences and Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Sequence limits**

Determine whether each sequence converges, and find the limit if it does:

a $a_n = \frac{3n+1}{n+2}$ Converges to 3 (divide by n).

b $a_n = \frac{n^2}{2^n}$ Converges to 0 (exponential growth dominates polynomial).

c $a_n = (-1)^n$ Diverges (oscillates between -1 and 1).

(a) Converges to 3 (divide by n).; (b) Converges to 0 (exponential growth dominates polynomial).; (c) Diverges (oscillates between -1 and 1).

2 **[Foundational]** • **Target: Series and partial sums**

Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.

Diverges — this is the harmonic series, a standard example of a divergent series with terms going to 0.

3 **[Foundational]** • **Target: Series and partial sums**

Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

$|r| = \frac{2}{5} < 1$: $S = \frac{3}{1-2/5} = 5$.

4 **[Foundational]** • **Target: Series and partial sums**

Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. The partial sums telescope: $S_N = 1 - \frac{1}{N+1} \to 1$.

5 **[Foundational]** • **Target: Monotonic and bounded sequences**

Show that the sequence $a_n = \frac{n}{n+1}$ is increasing, by comparing a_{n+1} and a_n.

$a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0$, so $a_{n+1} > a_n$: the sequence is increasing.

6 **[Foundational]** • **Target: Monotonic and bounded sequences**

Explain why the sequence $a_n = \frac{n}{n+1}$ (increasing and bounded above by 1) must converge, and find its limit.

By the Monotonic Convergence Theorem, a bounded, monotonic sequence converges. $\lim_{n\to\infty} \frac{n}{n+1} = 1$ (divide numerator and denominator by n).

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: More series practice

Evaluate the series $\sum_{n=1}^{\infty}\left[\left(\frac{1}{2}\right)^n+\left(\frac{1}{3}\right)^n\right]$ by splitting into two geometric series.

$\sum\left(\frac{1}{2}\right)^n=\frac{1/2}{1-1/2}=1$. $\sum\left(\frac{1}{3}\right)^n=\frac{1/3}{1-1/3}=\frac{1}{2}$. Total $=1+\frac{1}{2}=\frac{3}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More series practice

A ball dropped from 10 m rebounds to 60% of its previous height each bounce. Using an infinite geometric series, find the total distance traveled.

Total $=10+2(10(0.6)+10(0.6)^2+\cdots)=10+2\cdot\frac{6}{1-0.6}=10+2(15)=40$ m.

9 [Intermediate] • Target: Recursive sequences

A sequence is defined recursively by $a_1=2$, $a_{n+1}=\frac{a_n+6}{2}$. Compute a_2, a_3, and a_4.

$a_2=\frac{2+6}{2}=4$. $a_3=\frac{4+6}{2}=5$. $a_4=\frac{5+6}{2}=5.5$.

10 [Intermediate] • Target: Recursive sequences

Assuming the recursive sequence above converges to a limit L, find L by setting $L=\frac{L+6}{2}$ and solving.

$2L=L+6$, so $L=6$.

11 [Intermediate] • Target: Repeating decimals as series

Express the repeating decimal $0.\overline{45}$ as a fraction using an infinite geometric series with $a=\frac{45}{100}$ and $r=\frac{1}{100}$.

$S=\frac{45/100}{1-1/100}=\frac{45/100}{99/100}=\frac{45}{99}=\frac{5}{11}$.

12 [Intermediate] • Target: Visualizing partial sums of a series

The graph shows the first six partial sums $S_n=\sum_{k=1}^{n}\left(\frac{1}{2}\right)^k$ of a geometric series. Use the pattern to conjecture the value the sums are approaching, and confirm it using the geometric series formula.

The partial sums appear to approach 1. Using the geometric series formula: $S=\frac{1/2}{1-1/2}=1$, confirming the visual pattern.

13 [Intermediate] • Target: Series with factorial and combined terms

Evaluate $\sum_{n=0}^{\infty}\frac{1}{2^n}-\sum_{n=0}^{\infty}\frac{1}{3^n}$ by evaluating each geometric series separately.

$\sum\frac{1}{2^n}=\frac{1}{1-1/2}=2$. $\sum\frac{1}{3^n}=\frac{1}{1-1/3}=\frac{3}{2}$. Difference $=2-\frac{3}{2}=\frac{1}{2}$.

© ClipSAT — clipsat.org | All Rights Reserved

14 [Intermediate] • Target: Series with factorial and combined terms

A sequence is defined by $a_n = \frac{2n + (-1)^n}{n}$. Find $\lim_{n\to\infty} a_n$, explaining why the $(-1)^n$ term does not affect the limit.

$a_n = 2 + \frac{(-1)^n}{n}$. As $n \to \infty$, $\frac{(-1)^n}{n} \to 0$ (bounded numerator divided by an unbounded denominator), so $\lim_{n\to\infty} a_n = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 6.02 Optimization and Curve Sketching

Use the squeeze theorem to find $\lim_{n\to\infty} \frac{\sin n}{n}$, using the fact that $-1 \le \sin n \le 1$ for all n.

$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$ for $n \ge 1$. Since $\lim_{n\to\infty} \left(\pm\frac{1}{n}\right) = 0$, by the squeeze theorem $\lim_{n\to\infty} \frac{\sin n}{n} = 0$.

16 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 7.01 Integration by Substitution

Use the squeeze theorem to find $\lim_{n\to\infty} \frac{n!}{n^n}$, using the bound $0 < \frac{n!}{n^n} \le \frac{1}{n}$ for $n \ge 1$.

Since $0 \le \frac{n!}{n^n} \le \frac{1}{n}$ and $\lim_{n\to\infty} \frac{1}{n} = 0$, by the squeeze theorem $\lim_{n\to\infty} \frac{n!}{n^n} = 0$.

17 [Challenging] • Target: Applications of infinite series • Cross-Lesson: 7.02 Integration by Parts

A patient takes a 100 mg dose of medication every 8 hours. Between doses, 75% of the drug remaining is eliminated, so a fraction $r = 0.25$ carries over to just before the next dose. Using an infinite geometric series, find the long-run steady-state amount in the body immediately after a dose (assume doses accumulate indefinitely).

Immediately after the nth dose, the amount is $100 + 100r + 100r^2 + \cdots = \frac{100}{1 - 0.25} = \frac{100}{0.75} \approx 133.33$ mg in the long run.

18 [Challenging] • Target: Series with factorial and combined terms • Cross-Lesson: 8.01 Area Between Curves

Level 1 · Foundations Find the exact sum of the infinite geometric series: $\sum_{n=1}^{\infty} 3\left(-\frac{2}{5}\right)^n$

Identify the first term and ratio: When $n = 1$, $a = 3\left(-\frac{2}{5}\right)^1 = -\frac{6}{5}$. The common ratio is $r = -\frac{2}{5}$. Since $|r| = \frac{2}{5} < 1$, the series converges. Using the sum formula $S = \frac{a}{1 - r}$: $S = \frac{-6/5}{1 - (-2/5)} = \frac{-6/5}{7/5} = -\frac{6}{7}$

19 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 8.02 Volumes of Revolution

[Sequences & Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Applications of infinite series • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Level 1 · FoundationsVerify that the Mean Value Theorem applies to $f(x) = x + \frac{1}{x}$ on $[1, 4]$, and find all values of c that satisfy the theorem.

Step 1 · Check Hypotheses: $f(x)$ is continuous on $[1, 4]$ and differentiable on $(1, 4)$ since the only discontinuity $x = 0$ lies outside $[1, 4]$. The hypotheses of MVT hold. Step 2 · Compute Average Rate of Change: $\frac{f(4) - f(1)}{4 - 1} = \frac{(4 + 1/4) - (1 + 1)}{3} = \frac{17/4 - 2}{3} = \frac{9/4}{3} = \frac{3}{4}$. Step 3 · Solve $f'(c) = \frac{3}{4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Sequence limits

The sequence of partial sums $S_n = 2 - 1/2^n$: this means $\sum a_n =$

a. 1

b. 3/2

c. 2 ✓

d. diverges

Applying Sequences and Series principles, $\lim S_n = 2$. Therefore, the verified correct option is 2.

22 [Foundational] • Target: Series and partial sums

A monotone bounded sequence:

a. diverges

b. **converges ✓**

c. oscillates

d. is arithmetic

Applying Sequences and Series principles, Monotone convergence theorem. Therefore, the verified correct option is converges.

23 [Foundational] • Target: Monotonic and bounded sequences

An increasing sequence bounded above:

a. diverges

b. oscillates

c. **converges ✓**

d. is arithmetic

Applying Sequences and Series principles, Monotone convergence theorem. Therefore, the verified correct option is converges.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: More series practice

Squeeze theorem for sequences: if $a_n \le b_n \le c_n$ and $a_n, c_n \to L$, then $b_n \to$

a. L ✓

b. $L/2$

c. 0

d. $2L$

Applying Sequences and Series principles, Squeeze theorem. Therefore, the verified correct option is L.

25 [Foundational] • Target: Recursive sequences

Dirichlet test: $\sum a_n b_n$ converges if a_n monotone $\to 0$ and partial sums of $\sum b_n$ are:

a. convergent

b. zero

c. increasing

d. **bounded** ✓

Dirichlet: bounded partial sums + monotone $a_n \to 0$.

26 [Foundational] • Target: Repeating decimals as series

A sequence is bounded above if there exists M such that $a_n \le M$ for all n. This bound is:

a. always 0

b. the limit

c. **not necessarily unique** ✓

d. equal to the supremum

The bound M is not unique (any larger M also works).

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Visualizing partial sums of a series

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.

a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$ ✓

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_0^4((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_1^4((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

The region is bounded by $y = x^2$ and $y = 4$. To find the intersection points, we set $x^2 = 4$, which gives $x = \pm 2$. These are our limits of integration, $a = -2$ and $b = 2$. The region is revolved around the line $y = 5$. The outer radius $R(x)$ is the distance from the axis of revolution ($y = 5$) to the outer boundary of the region. The outer boundary is the curve $y = x^2$. So, $R(x) = 5 - x^2$. The inner radius of the cylindrical hole is given as r. The volume of the material removed by the cylindrical.

28 [Foundational] • Target: Series with factorial and combined terms

Consider the sequence defined by $a_1 = 1$ and $a_{n+1} = \frac{a_n^2 + k}{2}$ for $n \geq 1$, where k is a positive constant. If the sequence converges to a limit L, prove that $L^2 - 2L + k = 0$. Furthermore, if the limit L is an integer and $k = 3$, determine the exact value of the limit.

a. L = 3 ✓

b. L = 1

c. L = -1

d. $L = \sqrt{3}$

If the sequence converges to a limit L, then as $n \to \infty$, $a_{n+1} \to L$ and $a_n \to L$. Substituting this into the recurrence relation, we get: $L = \frac{L^2 + k}{2}$ Multiplying by 2: $2L = L^2 + k$ Rearranging the terms to form a quadratic equation: $L^2 - 2L + k = 0$ This proves the first part of the problem. For the second part, we are given that $k = 3$ and the limit L is an integer. Substituting $k = 3$ into the equation derived above: $L^2 - 2L + 3 = 0$ We can solve this qua.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: The Squeeze Theorem for sequences

Consider the region R bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 0$ and $x = 1$. A solid is formed by rotating R about the y-axis. Determine the volume of this solid using the method of cylindrical shells. Express your answer in terms of π.

a. $\frac{\pi}{2}$

b. $\pi \ln(2)$

c. π ✓

d. $\frac{\pi}{4}$

The region R is defined by $y = \frac{1}{x^2+1}$, $y = 0$, $x = 0$, and $x = 1$. We are rotating this region about the y-axis. Using the method of cylindrical shells, the volume V is given by the integral: $V = \int_a^b 2\pi x h(x) dx$ where x is the radius of the cylindrical shell, and $h(x)$ is the height of the shell. In this case, the radius of a cylindrical shell at a distance x from the y-axis is simply x. The height of the shell is the value of the function $y = \frac{1}{x^2+1}$. The limits of integr.

30 [Intermediate] • Target: Applications of infinite series

Consider the arithmetic sequence with the first term $a_1 = 5$ and a common difference $d = 3$. What is the value of the 5th term, a_5?

a. **17** ✓

b. 20

c. 15

d. 23

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$, where a_1 is the first term and d is the common difference. Given $a_1 = 5$ and $d = 3$, we want to find the 5th term, a_5. Using the formula: $a_5 = a_1 + (5-1)d$ $a_5 = 5 + (4)(3)$ $a_5 = 5 + 12$ $a_5 = 17$ Therefore, the 5th term of the sequence is 17.

31 [Intermediate] • Target: Sequence limits

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{10000}$?

a. 8 ✓

b. 9

c. 7

d. 10000

The error after summing up to $n = 7$ is at most the next term, $\frac{1}{40320} < \frac{1}{10000}$, while stopping one term earlier leaves a bound $\frac{1}{5040} \geq \frac{1}{10000}$. So 8 terms are needed.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Series and partial sums

A sequence $\{a_n\}$ converges if $\lim_{n\to\infty} a_n$ exists and is:

a. **finite (a real number)** ✓

b. infinite

c. 0

d. positive

Applying Sequences and Series principles, Convergence means finite limit. Therefore, the verified correct option is finite (a real number).

33 [Intermediate] • Target: Monotonic and bounded sequences

Sum of first n terms of arithmetic series: $S_n =$

a. na_1

b. $a_1 + nd$

c. $n(a_n - a_1)$

d. $\frac{n}{2}(a_1 + a_n)$ ✓

Applying Sequences and Series principles, $S_n = n(a_1 + a_n)/2$. Therefore, the verified correct option is $\frac{n}{2}(a_1 + a_n)$.

34 [Intermediate] • Target: More series practice

Squeeze theorem: $-x^2 \le f(x) \le x^2$ near $x = 0$ implies $\lim_{x\to 0} f(x) =$

a. 0 ✓

b. ± 1

c. DNE

d. 1

Applying Limits and Continuity principles, Squeeze: both bounds $\to 0$. Therefore, the verified correct option is 0.

35 [Intermediate] • Target: Recursive sequences

Sum of infinite geometric series ($|r| < 1$): $\sum_{k=0}^{\infty} ar^k =$

a. $\frac{a}{r}$

b. $\frac{a}{1-r}$ ✓

c. $a(1 - r)$

d. $\frac{a}{1+r}$

Applying Sequences and Series principles, $S = a/(1 - r)$, $|r| < 1$. Therefore, the verified correct option is $\frac{a}{1-r}$.

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Repeating decimals as series

Partial sum $S_n = \frac{2n}{n+1}$. Sum of series:

a. 0

b. 1

c. 3

d. 2 ✓

Applying Sequences and Series principles, $\lim_{n\to\infty} S_n = 2$. Therefore, the verified correct option is 2.

37 [Intermediate] • Target: Visualizing partial sums of a series

Fibonacci sequence: $1, 1, 2, 3, 5, 8, 13, \ldots$. The recurrence is $a_{n+2} =$

a. $a_n + 1$

b. $2a_{n+1}$

c. $a_{n+1} + a_n$ ✓

d. $a_{n+1} - a_n$

Applying Sequences and Series principles, $a_{n+2} = a_{n+1} + a_n$. Therefore, the verified correct option is $a_{n+1} + a_n$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Series with factorial and combined terms • Cross-Lesson: 8.01 Area Between Curves

Mertens' theorem: if $\sum a_n = A$ (absolutely) and $\sum b_n = B$, then the Cauchy product series $\sum c_n$ converges to:

a. $A + B$

b. $A - B$

c. A/B

d. AB ✓

Applying Sequences and Series principles, Mertens: Cauchy product $= AB$. Therefore, the verified correct option is AB.

39 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 8.02 Volumes of Revolution

Sum of n terms of geometric $1 + 2 + 4 + \cdots$: $S_n =$

a. $2^n - 1$ ✓

b. 2^n

c. $n \cdot 2^n$

d. 2^{n+1}

Applying Sequences and Series principles, $S_n = (1 - 2^n)/(1 - 2) = 2^n - 1$. Therefore, the verified correct option is $2^n - 1$.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Applications of infinite series • Cross-Lesson: 8.03 Arc Length and Improper Integrals

The sequence $a_n = \frac{n+1}{n}$ converges to:

a. 0

b. ∞

c. 2

d. 1 ✓

Applying Sequences and Series principles, $(n+1)/n = 1 + 1/n \to 1$. Therefore, the verified correct option is 1.

41 [Challenging] • Target: Sequence limits • Cross-Lesson: 9.01 Separable Differential Equations

Geometric sequence: $a_n = a_1 \cdot r^{n-1}$. If $a_1 = 2, r = 3$, then $a_4 =$

a. 6

b. 18

c. **54** ✓

d. 162

Applying Sequences and Series principles, $a_4 = 2 \cdot 27 = 54$. Therefore, the verified correct option is 54.

42 [Challenging] • Target: Series and partial sums • Cross-Lesson: 1.01 Precalculus Review for Calculus

Arithmetic sequence: common difference d. If $a_1 = 3, d = 4$, then $a_5 =$

a. 15

b. **19** ✓

c. 20

d. 23

Applying Sequences and Series principles, $a_5 = 3 + 4(4) = 19$. Therefore, the verified correct option is 19.

43 [Challenging] • Target: Monotonic and bounded sequences • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Find $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ using partial fractions:

a. 1

b. 1/2

c. 2/3

d. 3/4 ✓

$\frac{1}{n(n+2)} = \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2})$; telescopes to $\frac{1}{2}(1 + 1/2) = 3/4$.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: More series practice • Cross-Lesson: 12.01 Convergence Tests

Squeeze Theorem: If $-x^2 \le f(x) \le x^2$ near $x = 0$, then $\lim_{x \to 0} f(x) =$

a. -1

b. **0 ✓**

c. 1

d. cannot determine

Both bounds $-x^2$ and x^2 approach 0; by Squeeze, $f(x) \to 0$.

45 [Challenging] • Target: Recursive sequences • Cross-Lesson: 12.02 Power Series and Taylor Series

The sequence $a_n = \frac{\sin n}{n}$: does $\sum a_n$ converge?

a. **Yes (by Dirichlet test) ✓**

b. No

c. Yes (absolutely)

d. Inconclusive

Dirichlet: $1/n \to 0$ monotone; partial sums of $\sin n$ bounded.

© ClipSAT — clipsat.org | All Rights Reserved